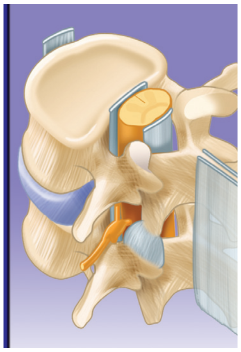

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Degenerative Conditions of the Cervical and Thoracolumbar Spine

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Degenerative conditions of the cervical, thoracic, and lumbar spine are encountered at all levels of medical referral. Because of their prevalence, all practitioners, from primary care providers to spine specialists, should possess a basic understanding of the epidemiology, pathophysiology, diagnosis, and treatment principles of these conditions. The aim of this chapter is to provide an overview of the conditions most frequently encountered by primary care and specialty physicians alike. With an understanding of these conditions, informed decisions may be made regarding their appropriate diagnosis, management, and specialty referral.

DEGENERATIVE CERVICAL SPINE

Axial Neck Pain

Cause and Epidemiology

Axial neck pain affects a significant portion of the US population. Its incidence increases with age, and multiple studies have shown a linear progression of its prevalence between the ages of 20 and 60. By the age of 65, as many as 95% of US adults suffer from some form of neck pain; in up to 10% of patients, the neck pain is severe.[1,2]

Numerous biomechanical and biochemical factors appear to contribute to the complex interplay that results in the sensation of axial neck pain. Examples include cumulative trauma, repetitive stress, and repetitive injury. Radiographic evidence of degenerative cervical spinal disease appears to contribute to neck pain, but the presence of such degeneration on imaging does not necessarily guarantee that a given patient will complain of neck pain.[3]

Cervical Motion Segment

The cervical motion segment is the basic functional unit of the cervical spine. It is composed of two adjacent cervical vertebral bodies and the intervertebral disc between them. Several soft tissue structures stabilize the cervical motion segment, including the anterior longitudinal ligament (ALL), posterior longitudinal ligament (PLL), interspinous ligament, ligamentum flavum, and the synovial capsules of the facet joints that flank the posterolateral aspect of the spinal canal bilaterally.[4]

The ALL is a strong band of the ligamentous tissue that runs along the anterior aspect of the vertebral bodies and intervertebral discs from the base of the skull to the sacrum. It functions as a primary stabilizer of the spine.[5] The PLL runs from the body of the C2 vertebra to the posterior surface of the sacrum. Its fibers coalesce with the ligamentum flavum anteriorly and the supraspinous ligament posteriorly. The multiple cervical motion segments work together to allow complex movement.[5]

Intervertebral Disks

The intervertebral discs are crucial to the normal function of the spinal unit. They are composed of a soft central nucleus pulposus, a tough, fibrous circumferential annulus fibrosus, and the two adjacent vertebral end plates on the proximal and distal aspects of the structure. Thus the disc unit provides a cushion between the two vertebrae it separates and facilitates motion between them. There are 6 cervical, 12 thoracic, and 5 lumbar intervertebral discs. The nucleus pulposus is composed of randomly oriented collagen fibrils and the proteoglycans that they bind. These molecules are hydrophilic, and the structure is composed primarily of water. Deforming forces on the structure cause it to disperse outward in various directions.[6] The annulus fibrosus is a tough, fibrous, circumferential structure. It is composed of layered collagen fibers organized in alternating directions that buffer the spinal column from complex multidirectional motions. The annulus prevents extrusion of the nucleus from between the vertebral end plates when compressive forces are applied. When prevention fails, such extrusion leads to the clinical phenomenon of a herniated intervertebral disc. The vertebral end plates are composed of hyaline cartilage and form a physical buffer between the intervertebral discs and vertebral bodies in the spinal column. Together the components of the intervertebral disc allow a complex host of movements, including axial compression, distraction, axial rotation, and lateral flexion.[6,7]

Facets

The spinal facet joints, otherwise known as apophyseal or zygapophyseal joints, are formed by the articulation between the superior articular facet of the inferior vertebra with the inferior articular facet of the superior vertebra; they are surrounded by robust synovial capsules. Each spinal motion segment contains two facet joints located between the pedicle and lamina of each respective vertebrae.[8] With their corresponding intervertebral discs, the facet joints serve a crucial role in the posterior and rotatory stability of the spinal column. As dictated by the three-column model of spinal stability, the spine is composed of three interrelated columns that form the overall structure. The anterior column is composed of the ALL and anterior two-thirds of the intervertebral discs. The middle column is composed of

Abstract

Degenerative conditions of the cervical and thoracolumbar spine are highly prevalent with advancing age. Commonly encountered conditions in the primary care and sports medicine settings that affect the cervical spine include axial neck pain, cervical spondylosis, discogenic pain, cervical disc herniation, foraminal stenosis, cervical radiculopathy, central stenosis, and cervical myelopathy. Rheumatoid cervical spondylitis and ossification of the posterior longitudinal ligament are additional cervical degenerative conditions that are frequently treated in these settings. Although similar forms of each of these disease entities also occur in the thoracic spine, they are relatively rare. Therefore the principal degenerative condition of clinical significance in this region is herniation of a thoracic disc. Low back pain is prevalent in the aging population and is encountered at all levels of referral. Other commonly encountered degenerative conditions affecting the lumbar spine include discogenic low back pain, lumbar disc herniation and radiculopathy, synovial facet cysts, degenerative lumbar spinal stenosis, and degenerative spondylolisthesis. With an understanding of degenerative spinal conditions, the clinician may accurately determine whether conservative management is indicated or whether a condition will be more amenable to referral to a spine specialist. Diagnosis hinges on a thorough history and physical examination as well as x-ray imaging. Advanced imaging such as computed tomography (CT) or magnetic resonance imaging (MRI) is typically mandatory to aid in diagnosis and, if indicated, operative planning. Although there are clinical nuances to the treatment of each of the discussed degenerative conditions, a basic treatment algorithm includes a trial of nonoperative therapy followed by operative management if symptoms fail to improve.

Keywords

cervical
thoracolumbar
stenosis
myelopathy
radiculopathy
herniation
spondylolisthesis

the posterior third of the vertebral body and disc and the PLL. The posterior column is composed of all structures posterior to the PLL. Facet joint orientation gradually transitions from coronal to sagittal alignment with progression towards the lumbar spine. Cervical facet joints are oriented at 45-degree angles in relation to the coronal plane and allow flexion, extension, lateral flexion, and rotation. Thoracic facet joints are situated at 60 degrees from the coronal plane and allow only lateral flexion and rotation. Lumbar facet joints are situated perpendicular to the coronal plane and thus allow only flexion and extension.[8]

Uncovertebral Joints

The uncovertebral joints, or Luscka joints, are found in the cervical spine between the C3 and C7 vertebrae. These joints form as a consequence of the normal degenerative changes associated with aging. The uncinate processes are superior projections found on the lateral aspect of the vertebral bodies.[9] As intervertebral joint space narrowing occurs with age, the lateral aspect of the proximal vertebral body gradually comes into contact with its corresponding uncinate process and the joint is formed. These joints are thought to serve multiple functions.[9] Because of their posterolateral location relative to the vertebral body, they provide a buffer in this region and protect adjacent structures that are at risk for disc herniation. Through a similar effect, they prevent posterior translation of the proximal vertebra relative to the distal vertebra and the associated neural injury that could ensue were this to occur.[9] Osteophytes that form as a result of uncovertebral arthrosis can have clinical implications secondary to compression of the cervical spinal nerve roots and vertebral arteries.[9]

Cervical Spondylosis

Epidemiology

Disc degeneration in the region of the cervical spine is referred to as cervical spondylosis, and it is a prevalent cause of neck pain in the aging population. It ultimately leads to a degenerative cascade that can compromise the space within the spinal canal and intervertebral foramina, and age is a strong predictor of the condition. Some 90% of men over 50 and women over 60 have evidence of cervical spinal degenerative changes on radiographic imaging (Figs. 130.1 and 130.2).[10] There is an increase in the annual incidence of degenerative changes between the cervical spinal levels of C3 and C7 up to age 50 as well as hyperosteogenesis and spinal stenosis up to age 60. After these age points the annual incidence appears to decrease. Cervical spondylosis in younger populations is generally due to prior trauma to the cervical spine.[10]

Pathoanatomy

The development of cervical spondylosis progresses in a stepwise fashion. It arises from a characteristic degenerative cycle that begins with intervertebral disc degradation. This generally leads to joint degradation, which in turn causes ligamentous changes. They then give rise to deformity and resulting kyphosis secondary to loss of disc height, with resulting load transfer to the facet joints and uncovertebral joints. This load transfer precipitates the formation of uncinate spurs and facet joint arthrosis. The cumulative effect of these actions is compression of the spinal cord and exiting nerve roots.[11]

Fig. 130.1 Anteroposterior x-ray of cervical degenerative disc disease.

Fig. 130.2 Lateral x-ray of cervical degenerative disc disease.

Discogenic Pain

Discogenic pain is multifactorial. The sinuvertebral nerve, a branch of the ventral ramus, innervates the spinal meninges, intervertebral disc, and PLL, and facet joint arthrosis stimulates the dorsal primary ramus, serving as another source of pain.[12] The pain of cervical spondylosis is similar to that reported with infection or tumor but lacks concurrent constitutional symptoms such as fever, sweats, weight loss, and chills. The pain associated with infection or tumor also tends to intensify with time and lacks specific patient-reported aggravating or relieving factors.[13]

Degenerative disc disease causes pain that is worsened by physical activity and relieved by rest. It may wax and wane intermittently, but for the most part it does not become markedly worse or better over long time periods. Symptoms such as extreme unremitting pain, extremity weakness or clumsiness, and inability to perform activities of daily living are concerning for infection or tumor and should not be overlooked.[13]

Cervical Disc Herniation, Foraminal Stenosis, and Cervical Radiculopathy

There are three types of cervical disc herniation. In the first type, bulging nucleus, the annulus remains intact and the herniating nucleus pulposus merely forms an outpouching as it presses against the annulus. In the case of the extruded disc type, the nucleus pulposus herniates through the annulus but is prevented from further herniation by a competent PLL. In the sequestered type, the material of the pulposus herniates through both the annulus fibrosus and nucleus pulposus and is free within spinal canal. Classification of herniated nucleus pulposus (HNP) based on location includes central, posterolateral, and lateral disc herniation. Disc herniation can cause compression of exiting spinal nerve roots and leads to foraminal stenosis and resultant cervical radiculopathy.[18]

Stenosis within the intervertebral foramina, formed laterally by the confluence of the superior and inferior vertebrae in the cervical spine, produces a symptom complex referred to as radiculopathy. Radicular pain is commonly defined as shooting pain that travels down an affected upper or lower extremity with compression of a spinal nerve root. When present in the cervical spine, it is typically described as shooting pain that originates in the neck and travels down the arm. When the history of radicular pain is being obtained, it is crucial to determine the distribution of the pain relative to the spinal nerve root affected, as pain and weakness typically follow a dermatomal distribution.[19]

For assessment of degenerative disk disease of the spine and spinal cord or nerve root compression, x-ray and magnetic resonance imaging (MRI) are the studies of choice.[13] Although their visualization of soft tissues is limited, x-rays should always serve as the first-line imaging studies. MRI should then be obtained to better visualize the soft tissue structures associated with the spinal cord and nerve roots, and this is usually sufficient alone when operative management of degenerative cervical spinal pathology is being planned. Plain radiographs are the primary imaging studies obtained during the initial evaluation of acute symptoms, when the examiner is not necessarily considering surgery. Unless there is a history of trauma, tumor, or infection, they do not necessarily need to be ordered at the first onset of pain. Radiographs provide general information characteristic of or related to cervical spondylosis as well as indications regarding osteophyte formation, bone quality of the cervical vertebrae, and the presence of osteophytes.[13]

Computed tomography (CT) scanning of the cervical spine provides additional information regarding the bony architecture and offers better resolution of osseous structures and landmarks than MRI. It is not, however, typically required as part of an imaging workup in a patient who has failed nonsurgical management of radiculopathy.[20]

Electrophysiologic studies are used primarily in the differentiation of radicular neck pain from other neuropathies of the upper extremity. Peripheral neuropathy can exist independently or concurrently in the setting of radiculopathy and is best distinguished and characterized using these studies. In the setting of concurrent neck pain, conditions such as carpal tunnel syndrome, brachial plexitis, and ulnar neuropathy can also masquerade as radicular pain from foraminal stenosis. Electrophysiologic diagnostic studies can also aid in the diagnosis of a patient with nonspecific upper extremity complaints that do not fit clearly with findings demonstrated on other imaging studies or with expected clinical syndromes.[13]

Central Stenosis and Cervical Myelopathy

Central cervical spinal stenosis, or the narrowing of the canal through which the cervical spinal cord travels in the cervical spine, causes the clinical presentation of cervical myelopathy.[14] Central cervical stenosis can arise anteriorly through the disc, with space-occupying herniation, disc calcification encroaching upon the volume of the spinal canal, or the formation of osteophytes (Fig. 130.3). It can also arise posteriorly, with hypertrophy of the posterior longitudinal ligament or facet joints bilaterally. A combination of these phenomena, leading to circumferential narrowing and compromise of the spinal canal, also commonly occurs.[14]

Central spinal canal narrowing, if sufficiently severe, causes myelopathy. The most common myelopathic presenting symptoms include neck pain, occipital headache, extremity weakness, paresthesias, and/or clumsiness. Gait and urinary disturbances are late findings and less reliable.[15] On imaging, patients with cervical myelopathy often display bony spurs of the intervertebral disc and vertebral body end plate, kyphotic cervical spine deformity, loss of disc space height, and anterior compression secondary to

Fig. 130.3 Magnetic resonance image of cervical spinal stenosis.

Fig. 130.4 Lateral magnetic resonance image of cervical spinal stenosis.

Fig. 130.5 Computed tomography image of the cervical spine.

osteophytes.[15] These changes lead not only to narrowing of the spinal canal but also to mechanical compromise of the functional spinal unit. If the unit becomes sufficiently compromised, the resulting mechanical changes can lead to ischemia and further compromise of spinal cord function (Fig. 130.4).[15]

The utilization of imaging studies is central to the diagnosis of cervical myelopathy. In the presence of clinical signs suggestive of myelopathy, imaging studies generally demonstrate a narrowed, compressed spinal canal. Sagittal canal diameter less than 10 to 13 mm, a cord compression ratio less than 0.40, and a Torg ratio of less than 0.75 are all predictive of elevated risk for presence or development of cervical spondylotic myelopathy.[16,17]

Rheumatoid Cervical Spondylitis

Rheumatoid cervical spondylitis presents in the vast majority of patients with rheumatoid arthritis (RA) and involves three patterns of instability: atlantoaxial subluxation, basilar invagination, and subaxial subluxation. Rheumatoid cervical spondylitis presents with signs and symptoms that are similar to those found in cervical myelopathy, including neck pain and stiffness, occipital headaches, and a gradual onset of weakness and loss of sensation in the extremities.[21] Physical examination of the patient with rheumatoid cervical spondylitis reveals the presence of hyperreflexia, weakness of the upper and lower extremities, and ataxia.[21] Flexion-extension radiographs are the primary imaging study obtained in the diagnosis of rheumatoid cervical spondylitis. CT imaging is obtained if surgery is being considered so as to better delineate bony anatomy, and MRI is used to evaluate the degree of spinal cord compression and identify insult to the cord.[21] Rheumatoid cervical spondylitis is treated first with nonoperative pharmacologic therapy for RA. Recent advances in the pharmacologic management of RA have led to a decrease in the operative management of rheumatoid cervical spondylitis, but in the event that nonoperative management fails and surgical intervention becomes necessary, spinal decompression and stabilization is used to prevent further progression of neurologic deficits. Patients should be counseled prior to surgery that operative intervention may not reverse existing neurologic deficits that were present before the decompression operation was performed.[21]

Atlantoaxial subluxation is present in 50% to 80% of patients with RA and most commonly occurs as anterior subluxation of C1 on C2 (Fig. 130.5). It results when pannus formation between the dens and the ring of C1 causes destruction of the transverse ligament and dens, leading to subluxation. Posterior C1-C2 fusion is indicated if the anterior atlanto-dens interval, measured on lateral flexion-extension radiographs, is greater than 10 mm, the posterior atlanto-dens interval (space available for cord) is less than 14 mm, or there is the progressive myelopathy on clinical presentation. Occiput-C2 fusion of the posterior C1 arch is indicated when atlantoaxial subluxation is combined with basilar invagination. Resection of the C1 posterior arch may be required in this operation if the atlantoaxial subluxation is not reducible.[21]

Basilar invagination, otherwise known as superior migration of the odontoid, or SMO, occurs when the tip of the dens migrates superiorly through the foramen magnum due to erosion and bone loss between the occiput and C1/C2. It is present in 40% of RA patients and commonly presents in combination with fixed atlantoaxial subluxation. Operative fusion of C2 to the occiput is indicated with progressive cervical migration greater than 5 mm, neurologic compromise, or a cephalomedullary angle less than 135 degrees on MRI.[21]

Subaxial subluxation is present in 40% of patients with RA and, unlike atlantoaxial subluxation and basilar invagination, it commonly occurs at multiple levels within the cervical and lumbar spine. Subaxial subluxation occurs due to pannus formation and soft tissue instability of the facet joints and Luschka joints. Subluxation greater than 4 mm or 20% is often consistent with cord compression. Cervical height index, a measure of body height/width of less than 2.0 mm, is an extremely sensitive and specific value in the prediction of impending neurologic compromise. Operative posterior fusion and wiring is indicated with subaxial subluxation greater than 4 mm or 20% in the presence of persistent pain and progressive neurologic compromise.[21]

Ossified Posterior Longitudinal Ligament

Ossified posterior longitudinal ligament (OPLL) is multifactorial in etiology. It commonly occurs in patients of Southeast Asian descent, affects men more than women, and most commonly presents between C4 and C6 in the cervical spine. Factors thought to contribute to the condition include diabetes, obesity, a high-salt diet, poor calcium absorption, and mechanical stress placed on the PLL. OPLL is often asymptomatic but can commonly present with symptoms of dysphasia. Less commonly, symptoms of myelopathy may be present. Radiographs of the lateral cervical spine demonstrate OPLL, MRI demonstrates soft tissue anatomy and cord compromise, and CT demonstrates features of the cervical bony anatomy (Fig. 130.6). Nonoperative observation is indicated only in patients with mild symptoms or those who are too medically unstable to undergo surgery. In patients presenting with dynamic myelopathy, interbody fusion without decompression is indicated. In patients whose kyphotic cervical spine precludes posterior decompression, anterior corpectomy with possible OPLL resection is indicated. In patients with a lordotic spine, in which the spinal cord is allowed to separate itself from the anterior compression caused by the OPLL, posterior laminoplasty or laminectomy with fusion may serve to treat the condition.[22]

Fig. 130.6 Lateral computed tomography of an ossified posterior longitudinal ligament.

DEGENERATIVE THORACOLUMBAR SPINE

Degenerative Thoracic Spine

Anatomic features of the thoracic spine differ significantly from those found in the cervical spine. Vertebral body size increases progressively from T1 to T12. The diameter of the spinal canal varies considerably in this region and is most narrow at T2 and T10. The zygapophyseal joints of the thoracic spine are oriented at approximately 60 degrees from the coronal plane and allow some rotation and minimal flexion and extension to protect the ribs.[18] The pedicle walls in the thoracic spine are twice as thick medially as they are laterally; their length decreases from T1 to T4 and then increases with each more distal vertebra. The shortest pedicle, located on T4, has an average length of only 14 mm.[18,23] Thoracic spinal nerves exit the spinal column laterally through intervertebral foramina and supply sensory and motor innervation to the chest and abdominal walls; degenerative changes in this region are less likely to be clinically significant.[23]

As in the cervical spine, pathologic entities including discogenic pain, disc herniation, foraminal stenosis, radiculopathy, spinal stenosis, and myelopathy can all present in the thoracic spine as well. Owing to the presence of the chest wall and articulations with the ribs, however, the thoracic spine is inherently more stable, and the majority of these conditions present only very rarely. Of the foregoing list, only thoracic disc herniation is sufficiently clinically relevant to warrant discussion in this chapter. Although it is the most common degenerative condition of the thoracic spine, thoracic disc herniation is a somewhat unusual condition overall, accounting for only 1% of all HNP events. It most commonly presents between the fourth and sixth decades of life because, with continued desiccation, the aging disc becomes progressively less likely to herniate through the surrounding annulus fibrosis. The herniation usually involves levels of the thoracic spine from the middle region to the lower levels, with 75% of all thoracic disc herniations occurring between T8 and T12.[18]

Thoracic degenerative disc herniation has a variable presentation. Although patients most commonly report a history of axial back or chest pain, band-like thoracic or abdominal radicular pain along the course of the affected intercostal nerve is also frequently reported. With a herniation between the T2 and T5 vertebral spinal levels, arm pain is a presenting symptom. Bowel or bladder changes are present in 15% to 20% of patients with thoracic disc herniation. Symptoms of numbness, paresthesias, sensory changes, myelopathy, and paraparesis distal to the level of the lesion as well as sexual dysfunction have also been reported. Physical examination of the patient with thoracic disc herniation reveals localized midline and paraspinal tenderness, paresthesias, and dysesthesia in the affected dermatomal distribution. With resultant cord compression, symptoms of myelopathy such as weakness, hyperreflexia, clonus, and presence of a Babinski sign can be noted.[18]

Diagnostic imaging of the affected spinal region is essential. Lateral thoracic spinal radiographs may be significant for disc space narrowing and osteophyte formation, but MRI more clearly delineates the soft tissue anatomy and is thus a more useful method of imaging thoracic degenerative disc herniation. The disadvantage of MRI scanning is, ironically, its image resolution. Because of MRI's high resolution, there is an inherently high false-positive rate as well. When the majority of asymptomatic patients are imaged using MRI, thoracic disc abnormalities are noted, with frank disc herniations and cord compression present in a smaller but significant portion of those imaged. Although these findings are most certainly structurally accurate, the imaging findings are not clinically relevant in the absence of corroborating symptoms as described previously.[24,25]

As with all cases of disc herniation, regardless of presenting spinal level, a trial of nonoperative therapy is first indicated. Interventions such as activity modification and physical therapy, as well as immobilization, short-term rest, analgesia, and thoracic spinal injections may be employed; these are successful in relieving symptoms in the majority of cases. Operative discectomy, which may include possible hemicorpectomy or fusion, is indicated in the setting of an acute disc herniation and in severe cases causing stenosis severe enough to lead to myelopathy on clinical history and physical exam. In contrast to myelopathy originating from the cervical spinal levels, myelopathic signs caused by thoracic spinal pathology appear only in the lower extremities. If there is obvious progressive neurologic deterioration, the need for surgery becomes much more pressing.[18]

Degenerative Lumbar Spine

Overview of Low Back Pain

Low back pain is an extremely common problem, with a lifetime prevalence of between 50% and 80% frequently reported in the literature. Because of this, the economic and social burden of the condition in developed countries is enormous. However, despite its prevalence, the true etiologic and pathogenic factors contributing to low back pain, especially mechanical low back pain, remain poorly understood. The most common cause of low back pain is a muscle strain.[26] Other common causes include overactivity, which can cause stretching and microinjury of muscular and ligamentous fibers, or acute injury to the intervertebral discs. Risk factors for low back pain are numerous and highly variable, including obesity and smoking, lifestyle factors such as prolonged or heavy lifting or prolonged sitting, or the presence of job dissatisfaction in the clinical history.[26,27] The astute clinician should not miss red flags for a more serious cause of the condition. The use of intravenous drugs with a history of fever and chills is concerning for an infectious etiology.[27] A prior diagnosis of cancer elsewhere in the body can be indicative of vertebral metastatic disease. A history of trauma, such as a motor vehicle accident or a fall from significant height, should arouse clinical suspicion for acute fracture and prompt the appropriate diagnostic workup. Finally, a history of bowel and bladder symptoms in conjunction with numbness and tingling in the saddle region should lead to prompt evaluation with appropriate imaging modalities and subsequent decompressive surgery if the condition is diagnosed. Despite these serious causes of low back pain, the vast majority of cases are caused by non–life threatening mechanical factors, and resolve without formal treatment within 1 year.

Lumbar Spine Anatomy

The anatomy of the lumbar region of the spine differs from that of the cervical and thoracic regions. In its normal nonpathologic alignment, an average of 60 degrees of lumbar lordosis is observed, ranging from approximately 20 to 80 degrees. The disc spaces account for the majority of this lordosis, and the L3 vertebra is the most prominent anteriorly in the lordotic curve.[28,29] The lumbar vertebral bodies are the largest in the axial spine and are accompanied by posterior arches; these are formed by pedicles and lamina, spinous processes, transverse processes, and the pars interticularis. The pars is a mass of bone unique to the superior and inferior articular facets and is the site at which spondylolysis occurs. The lumbar vertebral bodies receive their blood supply from segmental arteries, the dorsal branches of which also supply blood to the dura and posterior elements.[28,29]

Unlike those of the cervical spine, lumbar spinal nerve roots are matched to their corresponding vertebrae. Whereas a mismatch exists in the cervical spine (C6 nerve root travels under the C5 pedicle), in the lumbar spine the numbered nerve roots are matched and travel under their respective pedicles (L5 nerve root under L5 pedicle). The orientation of the nerve roots also serves to affect the clinical presentation produced when nerve roots are compressed by disc herniation.[30] Whereas the horizontal orientation of exiting cervical nerve roots leads to identical presentations with paracentral and foraminal disc herniations, the vertical orientation of lumbar spinal nerve roots produces distinct presentations depending on how far from the midline the disc herniation occurs. The spinal cord terminates in the filum terminale at approximately the level of T12; thus there is no true cord below this level. Neural impulses are instead transmitted via the cauda equina, which begins at L1 and traverses the remaining length of the spinal canal. Compression of these nerve roots, termed *cauda equina syndrome*, is a potentially devastating condition that, when discovered, warrants emergent surgical decompression and stabilization. Presenting symptoms of cauda equina syndrome include bilateral leg pain, saddle anesthesia, lower extremity sensorimotor changes, and bowel/bladder dysfunction.[30]

Discogenic Low Back Pain

Lumbar intervertebral discs are the main source of discogenic spinal pain in humans. Discogenic low back pain is a controversial condition that has only recently begun to become more widely accepted as a cause of idiopathic axial back pain. It is thought to originate from injury and resultant repair of the annulus fibrosus, and pathologic examination of discs of patients with discogenic low back pain demonstrates fissures in the annulus as well as abundant granulation tissue within these fissures that is extensively innervated.[26] Contrary to other tissue types, because of their avascular nature, the healing of intervertebral discs proceeds from the outside to the inside. With the exception of its posterior region, the normal disc is poorly innervated.[26] The posterior region of the disc receives innervation from both the sinuvertebral nerve and direct branches from the ramus communicans or the ventral ramus. Branches from the gray ramus communicans supply the lateral regions of the disc, and the anterior region is supplied by anterior discal nerves arising from the sympathetic plexus, which supplies the ALL. Sympathetic afferent fibers convey pain impulses.[26,27] A phenomenon referred to as *peripheral sensitization* occurs when mechanical stimuli that normally exert little effect on disc nociceptors can, under certain circumstances, generate a pain response that is amplified. Because of this, some damaged discs cause pain clinically and some do not.

The diagnosis of discogenic low back pain is based on clinical history and imaging modalities, which include x-ray, MRI, and provocative discography. Only about 20% of patients with low back pain have a defined pathologic or anatomic etiology that can be definitively identified. Lumbar spine radiographs are generally the first diagnostic imaging studies ordered; they may show findings consistent with disc degeneration and disc space narrowing. MRI demonstrates these findings in greater detail, showing degenerative discs without findings of stenosis or herniation.[31] Studies have shown that although provocative discography can aid in making the diagnosis, it can lead to the acceleration of disc degeneration as an increased likelihood of disc herniation, changes in the vertebral end plates, and loss of disc height over time.[32]

A trial of nonoperative management is indicated as a first-line therapeutic modality in the treatment of discogenic low back pain. Pharmacologic treatment, including analgesics, nonsteroidal antiinflammatory drugs (NSAIDs), and muscle relaxants are commonly employed, but evidence for their efficacy is lacking. Long-term treatment with narcotics is generally avoided, although there is limited evidence of its ability to allow for a small improvement in pain and function.[26] Exercise therapy via the McKenzie method is popular among physical therapists and has been shown to be slightly more effective than manipulation and as effective as strength training. If pharmacologic therapy, lifestyle modification, and exercise fail to improve symptoms, epidural steroid injections are commonly performed. Although there has been significant debate regarding the effectiveness of these injections and indication for their use, recent systematic reviews have shown that the alleviation of pain and functional relief of symptoms experienced by patients who received them were at least fair.[26] Although their mechanism of action is not completely understood, it is believed to be secondary to the generation of a neural blockade that interrupts nociceptive input, the reflexive mechanism of the afferent fibers, and the self-sustaining central pattern of pain-generating neurons. Corticosteroids also reduce inflammation by inhibiting the release of numerous inflammatory mediators and in doing so produce a local anesthetic effect. Although some authors advocate lumbar spinal fusion as a potential option for refractory discogenic low back pain, this procedure should be approached with caution, as the reported results vary considerably among studies and the rate of complications following these operations is not negligible.[26]

Lumbar Disc Herniation and Radiculopathy

The topic of lumbar disc herniation and resultant radiculopathy is enormous and large enough to compose an entire chapter or even textbook alone. The purpose of this section is not to provide an exhaustive review of the topic but rather to outline major principles of the pathophysiology, epidemiology, diagnosis, and treatment of the condition. As in the cervical and thoracolumbar regions of the spine, lumbar disc herniation occurs when the central region of the intervertebral disc, the nucleus pulposus, extrudes through the annulus fibrosus. Resultant irritation and compression of the dural sac or lumbar nerve roots leads to the clinical symptoms of lumbar radiculopathy, which is the most common reason for the performance of spinal surgery. Moreover, rates of surgical intervention have increased owing to the higher prevalence of advanced imaging and the improved safety of surgical procedures. Disc herniation occurs in all age groups but is most commonly reported in the fourth and fifth decades of life. No obvious risk factors for the development of lumbar radiculopathy have been identified, and although smoking and bearing repetitive and vibratory loads both lead to an increase in incidence of the condition, the difference they produce is small.[33]

The sciatic pain of lumbar radiculopathy is multifactorial but probably most closely related to mechanical compression of the lumbar nerve roots as a result of disc herniation, which occurs secondary to degenerative changes within the involved structures, thus weakening them and predisposing them to failure. The resultant ischemia sensitizes the nerve roots and their dural coverings to pain, and symptoms of sciatica are generated.[34] Extruded disc herniations, in which the material of the pulposus pierces through the annulus, produce higher levels of inflammatory cell infiltration. These inflammatory cells are thought to originate from the peripheral vasculature and are important for the generation of sciatic pain. Since this phenomenon is less apparent in nonextruded (contained) disc herniation, the predominant effect in contained herniation is that of simple mechanical compression. In extruded herniation, the dominant effect is that of inflammatory cell activity.[34,35]

As with most conditions of the lumbar spine, the clinical presentation of lumbar disc herniation is variable. The typical presentation begins as an initial period of low back pain that progresses to leg symptoms after approximately a week and sometimes persists further to only involve the legs. Since lumbar radiculopathy usually occurs secondary to a focal area of disc herniation, symptoms usually present unilaterally. Because of the somewhat more complex course of lumbar spinal nerves as they exit the canal, the position of lumbar disc herniation has implications for the symptom complex observed. Central (midline posterior) herniation is commonly associated only with back pain; it may, however, be associated with symptoms of cauda equina syndrome and constitute a surgical emergency.[36] Paracentral (posterolateral) herniation causes symptoms in the distribution of the distal spinal nerve root as it descends to its point of exit through the intervertebral foramen. For example, a herniation of the L4/L5 disc in this position causes symptoms in the L5 distribution. Conversely, far lateral (foraminal) herniation causes symptoms in the distribution of the proximal nerve root as it exits the foramen. In this scenario, a herniation of the L4/L5 disc causes symptoms in the L4 distribution. Physical examination must include thorough testing of all lumbar nerve root distributions for weakness and reflex abnormalities.[36]

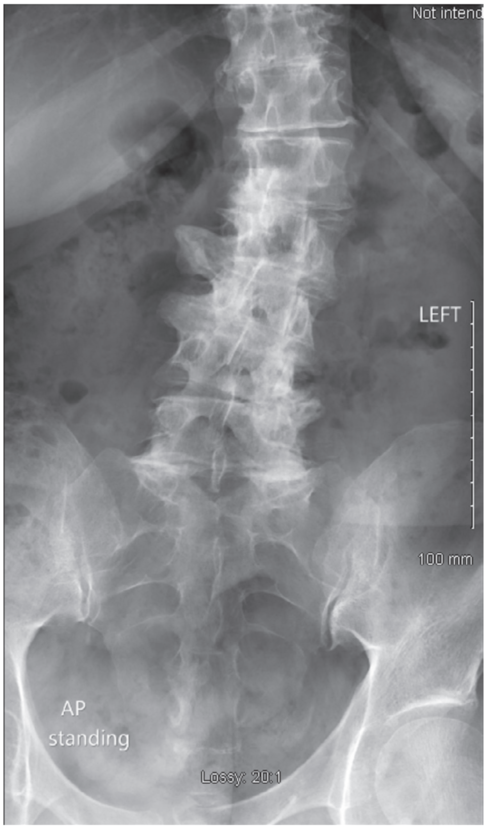

Fig. 130.7 Anteroposterior x-ray of lumbar degenerative joint disease.

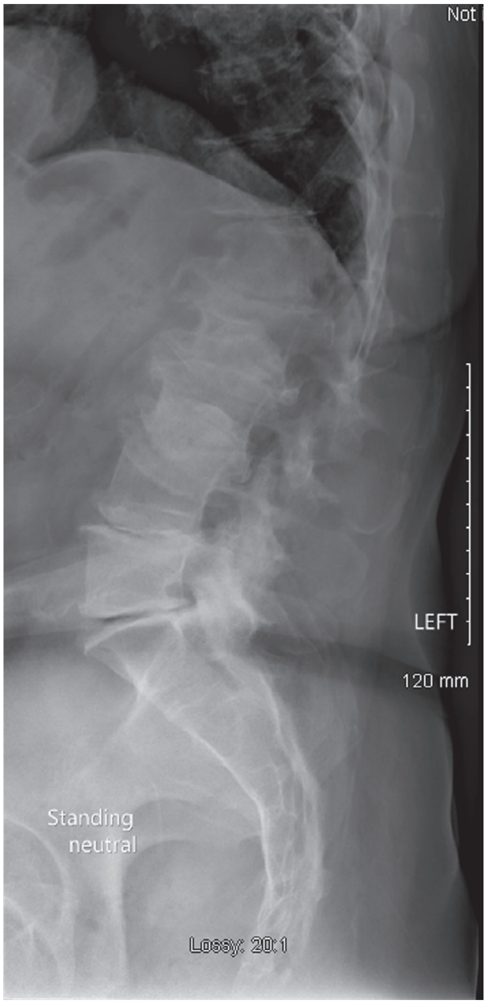

Fig. 130.8 Lateral x-ray of lumbar degenerative joint disease.

Since obtaining plain radiographs of the lumbar spine is inexpensive, radiography should be performed in all cases that require evaluation for the condition (Figs. 130.7 and 130.8). Although plain radiographs are not useful in evaluating the soft tissue elements affected by lumbar radiculopathy, they have utility in excluding other conditions that could be contributory. The primary imaging modality utilized in the evaluation of lumbar disc herniation is MRI, and the thorough assessment of the soft tissues it allows renders it indispensable in the diagnostic algorithm (Figs. 130.9 and 130.10). Lumbar disc herniations are classified on MRI according to shape. A disc herniation whose base is wider than its height from the margin of the disc is referred to as a *protrusion*, which may be focal or broad and concentric. A herniation whose base is narrower than its height is an *extrusion*. A herniation whose pulposus material has lost continuity with the disc is referred to as a *sequestration*.[37]

Because the natural history of lumbar disc herniation includes relief of symptoms over a period of 4 to 6 weeks, with 90% of patients' symptoms resolving by 3 months, the mainstay of initial therapy is conservative treatment. Conservative therapeutic modalities focus on relieving pain, increasing physical activity levels, and avoiding rest. NSAIDs are the primary medical therapy administered owing to their antiinflammatory activity. Analgesics may also be used to assist the activity of NSAIDs if patients report persistent pain.[38] Transforaminal injection of anesthetics and corticosteroids has been shown to be a safe and effective alternative to oral therapy secondary to lumbar disc herniation. In addition to analgesic and antiinflammatory medications, physiotherapy that includes gentle exercise and stretching is essential to the conservative treatment algorithm.[39]

Although most cases of lumbar disc herniation with resultant radiculopathy resolve spontaneously with conservative therapy alone, absolute and relative indications for operative management of the condition do exist. As with many conditions affecting the spinal cord, absolute indications for urgent surgical management include symptoms of cauda equina syndrome or substantial paresis on exam.[36] Relative indications for operative management include sciatica that is unresponsive to conservative therapy for at least 6 weeks or motor deficit greater than grade 3 on physical examination. Recently advocates for early surgical intervention in the literature have cited the faster return to function despite equivalent outcomes of operative management of lumbar radiculopathy and the economic favorability of surgical management due to faster return to work. Operative management of lumbar radiculopathy generally involves some form of a discectomy. Although traditional discectomy is still utilized by some surgeons, microdiscectomy and minimally invasive procedures have gained favor in recent years. Owing to the favorable results associated with microdiscectomy, both during the operation and in the long term, it is the preferred surgical technique. Aggressive removal of lumbar disc material

Fig. 130.9 Lumbar herniation, sagittal T2 image.

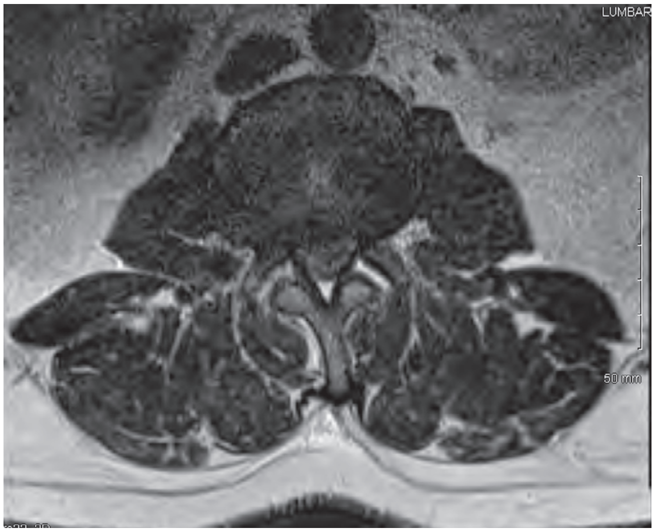

Fig. 130.10 Lumbar herniation, axial T2 image.

has been associated with a greater incidence of lumbar pain and greater acceleration of degenerative changes in the lumbar spine. Because the ligamentum flavum provides a barrier between the nerve roots, the dura mater, and the epidural fat, its preservation during the operation is thought to result in a favorable prognosis and effective formation of epidural fibrosis following the operation.[40] Arthrodesis or arthroplasty do not have a clear role in the typical operative management of disc herniation.[36]

Synovial Facet Cysts

The etiology of synovial facet cysts, cystic formations arising from the synovial lining of the posterolateral synovial joint capsules, remains unclear, but factors including spinal instability, facet joint arthropathy, and degenerative spondylolisthesis (DS) are associated with the formation of spinal cysts. Synovial facet cysts occur predominantly in the lumbar spine. Patients with lumbar synovial facet cysts generally present in their 60s, but such cysts have been discovered in patients ranging from 28 to 94 years of age. Female:male ratios of between 1 : 1 and 4 : 1 have been reported in the literature.[41] Although the epidural growth of synovial cysts in the spinal canal and the resulting compression of neural structure causes the clinical symptoms associated with synovial facet cysts, they may also be asymptomatic incidental findings. The clinical presentation of synovial facet cysts depends on their size and relation to surrounding neural structures. In patients who do present with clinical symptoms, low back pain is the most common symptom, followed by radicular pain, sensory deficits, motor deficits, and reflex abnormalities.[42] A small minority of patients present with cauda equina, lateral recess, or spinal stenosis syndrome. Synovial cysts are lined internally with cuboid or pseudostratified columnar epithelial cells and filled with clear or straw-colored fluid. They can rarely hemorrhage and bleed into surrounding soft tissues, causing mass effect and resultant spinal cord compression.[43]

The diagnosis of symptomatic synovial facet cysts is based on clinical presentation and imaging modalities. Plain radiographic imaging does not contribute significantly to the diagnosis of synovial facet cysts, but it is important in the exclusion of other conditions such as DS and metastatic spinal disease. CT and MRI are both used routinely to characterize synovial facet cysts and to plan for their operative management. On MRI, synovial cysts appear as smooth, well-circumscribed extradural cystic masses located in close proximity to the facet joints. Because of its excellent visualization of the soft tissue elements of the lumbar spine, MRI is the diagnostic imaging modality of choice in the workup of a presumed synovial facet cyst. The appearance of synovial cysts on CT is dependent on its content and factors such as calcification, blood, inflammatory activity, and osseous structure involvement can all affect the appearance of the cyst with this modality.[44]

Owing to its mobility, the L4/L5 vertebral level is the site of presentation of a majority of synovial facet cysts. L5-S1 is considered the second most common site of development of these cystic masses, followed by L3-L4 and L2-L3. The development of synovial cysts is independent of laterality, and cysts are found equally in left- and right-sided facet joints of the lumbar spine.[42] Numerous factors have been associated with the development of these cysts, including degenerative spondylosis, spinal instability, spinal trauma, facet arthropathy, and hypermobile facet joints. DS appears to be strongly associated.[42]

Management of synovial facet cysts begins with a trial of conservative therapy. Observation, oral analgesics, physical therapy, and bracing have all been reported in the literature. After failure of conservative modalities, CT-guided needle aspiration, cyst punctures, and intra-articular corticosteroid injection may be employed. Although direct puncture of cysts by means of translaminar CT can be an effective means of temporary relief, most cysts managed by this method will recur within 6 to 12 months.[45] Steroid injections, first reported in 1985, typically yield short-term improvement in symptoms or no improvement at all, but a few series have reported acceptable long-term results.[46] In cases of intractable pain or residual neurologic deficits, surgical treatment is indicated. The specific style of surgical management depends primarily on the location, size, and involvement of surrounding structures. Surgical management involves removal of the offending cyst followed by additional spinal fusion procedures as indicated for resultant or preexisting spinal instability, listhesis, or spondylosis.[47] Outcomes following surgical management of synovial facet cysts for intractable clinical symptoms have generally been favorable, and operative management of the condition must be tailored to each individual patient based on the clinical presentation and imaging findings.[47]

Degenerative Lumbar Spinal Stenosis

Degenerative lumbar spinal stenosis (LSS) is by definition a secondary, or acquired, stenosis resulting from the cumulative effect of lumbar degenerative changes. The canal stenosis associated with the condition may result from narrowing in multiple dimensions in the axial plane, including the anteroposterior, transverse, or combined diameter.[48] Structures whose pathology can contribute to degenerative LSS include the intervertebral disc, facet joints, and ligamentum flavum. Hypertrophy of the ligamentum flavum occurs secondary to cumulative mechanical stress, most notably along the structure's dorsal aspect. Decreased disc height and facet joint hypertrophy also contribute to this process, and the cumulative effect of these processes serves to narrow the central spinal canal.[48] In addition to the slow progression of these degenerative changes, acute changes in lumbar spinal position lead to changes in the volume available for the contained neural structures. With axial loading and extension in the sagittal plane, the space within the lumbar spinal canal decreases (Fig. 130.11). Conversely, with axial distraction and flexion in the sagittal plane, the space within the canal increases. The effects of these acute changes in position have important clinical implications in the evaluation of LSS, and questions regarding these effects should always be posed to the patient during his or her evaluation.[48]

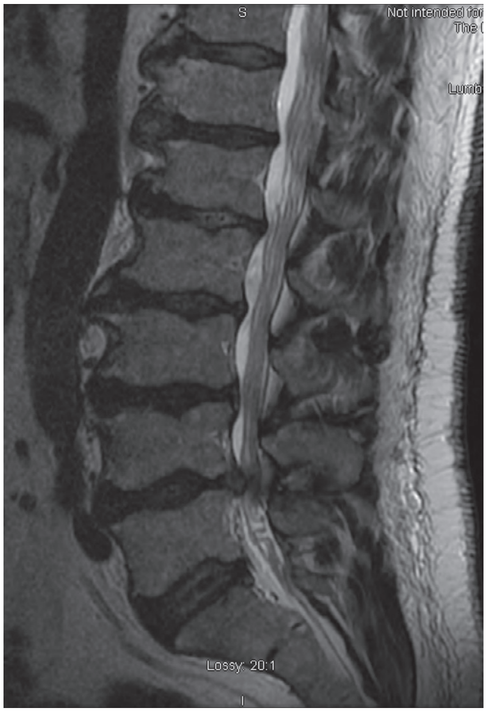

Fig. 130.11 Magnetic resonance image of lumbar spinal stenosis.

Evaluation of a patient with presumed LSS begins with the history and physical examination. Neurogenic claudication is most commonly reported; it causes shooting pain radiating from the low back to the buttock, posterior thigh, and posterior leg. Symptoms can also include involvement of the groin and anterior thigh as well as fatigue, heaviness, weakness, and paresthesias.[49] Symptoms are most commonly reported bilaterally but can less commonly present unilaterally, and leg pain is usually the most troublesome symptom.[49]

In neurogenic claudication, flexion increases available space within the spinal canal and reduces symptoms, whereas extension reduces space and increases symptoms. Patients will typically report exacerbation of symptoms with lying flat and alleviation with lying on their side or going up stairs, which flexes the spine. They walk with a typical "simian stance," which involves slight knee and hip flexion, allowing slight flexion of the spinal canal. In vascular claudication, spinal position has no effect on symptoms, and the distance patients can walk before developing symptoms is shorter. Descending stairs does not cause pain, and pulse examination is abnormal.[50]

Radiologic imaging is helpful in the diagnosis of degenerative LSS, but it is important that the findings be correlated with the clinical history. Although there is no universally accepted definition of central, lateral recess, and foraminal stenosis, the values most commonly cited are relative spinal stenosis at a diameter of 10 to 12 mm and absolute spinal stenosis at a diameter less than 10 mm.[51] Criticisms of this method include its inability to consider the trefoil shape of the spinal canal and intrusive behavior of the ligamentum flavum and disc material in degenerative spinal stenosis.[52] Another view, proposed by Schonstrom et al., is that constrictions in cross-sectional area that leave 70 to 80 mm^2 of the spinal canal's area unconstructed would be unlikely to cause clinical symptoms consistent with cauda equina syndrome. Any degree of stenosis, however, can lead to nonurgent symptoms of nerve root compression.[53]

The treatment of degenerative LSS begins conservatively, with treatment modalities including medications, physical therapy, exercises and bracing, and epidural injections. Although conservative treatment modalities are almost always employed as the first-line therapy for this condition, there is little evidence that in the literature to guide their utilization. Commonly used medications include analgesics, NSAIDs, muscle relaxants, and opioids, each posing unique risks and challenges to their administration.[48] A course of physical therapy with comprehensive rehabilitation including manual therapy, stretching, and strengthening of the lumbar spinal musculature and hips is frequently utilized, with an emphasis on endurance exercises. Although courses involving these modalities are extremely prevalent early in the treatment course of degenerative LSS, few prospective randomized trials have evaluated their effectiveness. When medical therapy, physical therapy, and endurance training fail to alleviate symptoms, epidural corticosteroid injections are commonly employed. Although

30% of epidural corticosteroid injections are performed for LSS, there are few data to support their efficacy.[48]

When conservative treatment modalities fail to relieve the pain and disability associated with degenerative LSS, surgery may be considered as a treatment option. Surgical management of LSS typically involves a decompressive laminectomy with or without concomitant fusion to address the instability thought by some surgeons to be generated during posterior element removal and canal decompression.[54] The long-term success rates of decompressive laminectomy are generally favorable, depending on the cited endpoint, ranging from 45% to 72%; two recent major studies found improvement in pain and pain/function for surgery versus nonoperative treatment, respectively.[54] When they compared unilateral laminotomy, bilateral laminotomy, and laminectomy, Thomé et al. found bilateral laminotomy to provide superior relief of back and leg pain both at rest and while walking. However, walking distance without symptom development was improved in all three groups.[55] Although serious complications and death are very rare with surgical management of degenerative LSS, they can occur. Thus as with any surgical procedure, the patient must be counseled thoroughly prior to consenting to the procedure. The true complication rate of the procedure remains largely unknown, since few large studies have reported complication rates associated with surgical management. Shared decision-making is an important component of preoperative planning, and patients must be active members of the process. Overall the outcomes of surgical management of degenerative LSS are excellent, with improvements in quality of life similar to those experienced by patients undergoing total knee arthroplasty.[54]

Degenerative Spondylolisthesis

DS is a condition in which degenerative changes in the lumbar spine cause a proximal vertebral body to slip anteriorly or posteriorly relative to the vertebral body distal to it.[56] Symptoms in DS generally develop in patients over the age of 40, and the condition causes a variable clinical presentation. In one study, a mean slip of 14% was reported, highlighting the mild degree to which DS causes anterior translation of the proximal vertebral body relative to the distal.[57] Because the neural arch remains intact, however, even a small degree of slip can generate a shear force that is substantial enough to cause cauda equina syndrome. Although the precise cause of DS is unknown, the factors most intimately related probably include (1) facet joint arthritis, leading to a loss in normal structural support; (2) stabilizing ligament malfunction, most likely secondary to hyperlaxity; and (3) poor muscular control, leading to a lack of effective secondary stabilization. These factors in concert are thought to lead to sagittal instability and resultant DS.[56] With progression of the slip, the facet joint capsules hypertrophy, the ligamentum flavum buckles, and the intervertebral discs bulge forward, further contributing to the slip. Risk factors for DS include age greater than 50, multiparity, sagittally oriented facet joints, African American ethnicity, hyperlordosis, and elevated pelvic incidence. DS occurs at L4/L5 at a rate 6 to 9 times greater than at other levels owing to the strength of the iliolumbar ligaments, which hold L5 firmly in place without stabilizing L4.[56]

As with many conditions affecting the lumbar spine, the main complaint of the patient with DS is back pain.[58] Patients typically describe the pain as intermittent and a recurring pattern of symptoms that may occur over many years. Pain is generally worse with physical activity or changes in position of the lumbar spine, and commonly worsens throughout the day. Symptoms in the legs also occur frequently and are commonly the reason patients ultimately become concerned and seek formal evaluation. Leg pain typically presents diffusely in the lower extremities, involving the distribution of the L5 and/or L4 nerve roots. A history of alternation of leg pain between the legs is a common complaint and strongly suggestive of DS.[58] When radiculopathy presents unilaterally, it is usually due to L5 compression at the level of the lateral recess. Extreme stenosis can result from DS; when it occurs, it can lead to symptoms of cauda equine syndrome. In contrast to the acute, sometimes devastating bowel and bladder symptoms caused by lumbar disc herniation, symptoms in DS are often insidious and subtle. As with all stenotic conditions of the lumbar spine, adoption of a position of flexion increases the anteroposterior diameter of the spinal canal and alleviates symptoms.[58]

The diagnosis of DS is confirmed with imaging studies. Although plain radiographs are typically sufficient to diagnose DS (Fig. 130.12), advanced imaging studies are required for preoperative planning. The essential finding on the lateral radiographic view is anterior displacement of L4 on L5 or less commonly L5 on S1 or L3 on L4, with an intact neural arch noted posteriorly.[59] Because of the intact pars interarticularis, spinous processes are visualized translated anteriorly with their

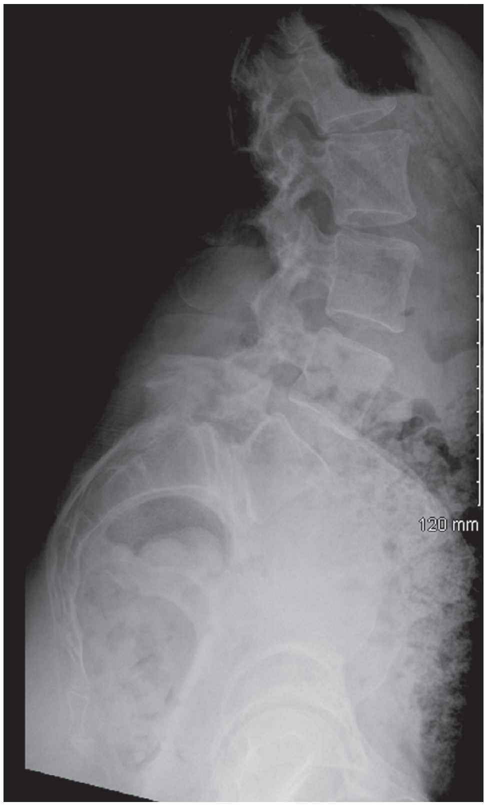

Fig. 130.12 Lumbar spondylolisthesis.

Fig. 130.13 Magnetic resonance image of lumbar spondylolisthesis.

Fig. 130.14 Magnetic resonance image of spondylolisthesis foraminal stenosis.

corresponding vertebral bodies. On the anteroposterior lumbar radiograph, hemisacralization of L5 is typically noted. Generalized findings of a degenerative lumbar spinal disease are also noted, including narrowing of the disc spaces, sclerosis of the vertebral end plates, marginal peridiscal osteophytes, and hypertrophy and sclerosis of the facet joints.[59] MRI further characterizes the osseous features of DS and also demonstrates the degree to which the neural and dural soft tissues are affected (Figs. 130.13 and 130.14). MRI should be ordered for diagnostic confirmation in all but the most obvious plain radiographic cases and should always be obtained when surgical management of DS is being considered.[59]

Two methods are used to grade spondylolisthesis. The first, proposed by Meyerding, divides the superior aspect of the distal vertebra into four quadrants and grades slips I to IV based on the position of the posterior cortex of the proximal vertebral body on lateral/sagittal imaging. The second method, proposed by Taillard, is similar but expresses the slip of the proximal vertebral body on the distal vertebral body as a percentage of the total anteroposterior diameter.[60] Taillard's method is more widely accepted because of its reproducibility.[59]

The treatment algorithm of DS is similar to that of most degenerative conditions affecting the lumbar spine. Conservative treatment can usually be trialed initially. Approximately three-quarters of patients who are neurologically intact on initial presentation do not deteriorate over time; if this is the case, conservative treatment is appropriate.[61] If patients describe a history of neurogenic claudication or have bowel or bladder symptoms on initial presentation, conservative therapy should be bypassed in favor of primary surgical management owing to poor outcomes when these patients are treated nonsurgically. Even in the presence of nonclaudicatory neurologic symptoms, a patient with low back pain and DS should be managed nonoperatively.[61] No optimal protocol has been definitively established in the literature, and although many have been proposed, most share the same basic elements. Vibert et al. described a typical protocol as 1 to 2 days of rest, a short course of antiinflammatory medications to follow, and referral to physical therapy if symptoms persist longer than 1 to 2 weeks.[62]

A complete discussion of the operative management of DS is beyond the scope of this chapter. In general, however, indications for surgical treatment of DS include (1) persistent/recurrent low back pain, leg pain, or neurogenic claudication, causing a significant diminution in quality of life and persisting despite at least 3 months of nonsurgical treatment; (2) a progressive neurologic deficit of any kind; and (3) bowel or bladder symptoms of any kind.[63] When any of these are present, surgical treatment is indicated and should be performed expeditiously. Operative management of DS generally involves posterior decompression with posterolateral fusion.[64] Reduction of the listhesis is generally deferred in favor of laminectomy, wide decompression, and foraminotomy.[64]

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